PROGRAM:

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#include <stdio.h>
#include <stdlib.h>
struct heap {
    int c;
 // capacity
    int s;
 // size
    int *element;
 };
typedef struct heap *pqueue;
pqueue initialize(int max) {
    pqueue heap1;
    if (max <= 3) {
        printf("\n Priority queue is too small\n");
        exit(EXIT_FAILURE);
    }
    heap1 = (pqueue)malloc(sizeof(struct heap));
    if (heap1 == NULL) {
        printf("\n Out of space\n");
        exit(EXIT_FAILURE);
    }
    heap1->element = (int *)malloc((max + 1) * sizeof(int));
    if (heap1->element == NULL) {
        printf("\n Out of space\n");
        exit(EXIT_FAILURE);
    }     heap1->c = max;
    heap1->s = 0;
    heap1->element[0] = 0;
    return heap1;
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 }
 int isempty(pqueue heap1) {
    return (heap1->s == 0);
 }
int isfull(pqueue heap1) {
    return (heap1->s == heap1->c);
 }
void insert(int x, pqueue heap1) {
    int i;
    if (isfull(heap1)) {
        printf("\n Heap is full\n");
        return;
    }
    for (i = ++heap1->s; heap1->element[i / 2] > x; i = i / 2)
        heap1->element[i] = heap1->element[i / 2];
    heap1->element[i] = x;
 }
int deletemin(pqueue heap1) {
    int i, child, min, last;
    if (isempty(heap1)) {
        printf("\n Heap is empty\n");
        return heap1->element[0];
    }
    min = heap1->element[1];
    last = heap1->element[heap1->s--];
    for (i = 1; i * 2 <= heap1->s; i = child) {
        child = i * 2;
        if (child != heap1->s && heap1->element[child + 1] < heap1->element[child])
            child++;
        if (last > heap1->element[child])
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            heap1->element[i] = heap1->element[child];
        else
            break;
    }
    heap1->element[i] = last;
    return min;
}
void display(pqueue heap1) {
    int i;
    printf("\nElements in the heap: ");
    for (i = 1; i <= heap1->s; i++) {
        printf("%d ", heap1->element[i]);
    }
    printf("\n");
}
int main() {
    int s, ch, ele;
    pqueue heap1;
    printf("\nEnter the size of the queue: ");
    scanf("%d", &s);
    heap1 = initialize(s);
    do {
        printf("\nMenu\n1. Insert\n2. Delete\n3. Display\n4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &ch);
        switch (ch) {
            case 1:
                printf("\nEnter the element to insert: ");
                scanf("%d", &ele);
                insert(ele, heap1);
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            break;
        case 2:
            ele = deletemin(heap1);
            printf("\nThe deleted element is %d\n", ele);
            break;
        case 3:
            display(heap1);
            break;
        case 4:
            printf("\nExiting...\n");
            break;
        default:
            printf("\nInvalid choice\n");
      }
   }
while (ch != 4);
   free(heap1->element);
   free(heap1);
   return 0;
}
```